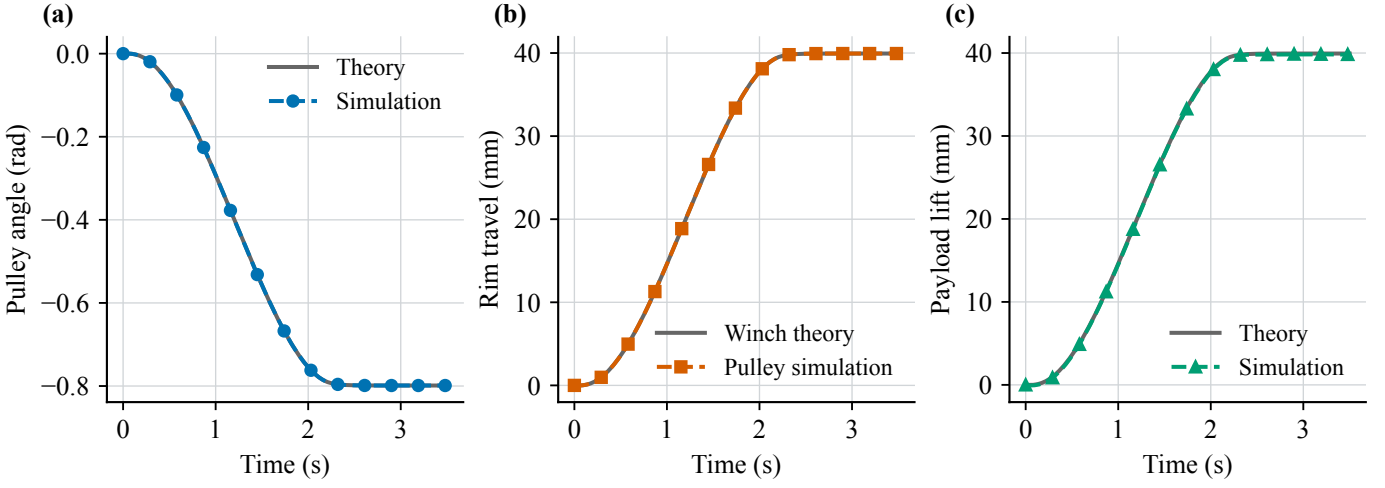
(a)
Pulley angle (rad)
0.0
−0.2
−0.4
−0.6
−0.8
Theory
Simulation
0
1
2
3
Time (s)
(b)
Rim travel (mm)
40
30
20
10
0
Winch theory
Pulley simulation
0
1
2
3
Time (s)
(c)
Payload lift (mm)
40
30
20
10
0
Theory
Simulation
0
1
2
3
Time (s)